

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Seventh Semester B.E. (E&E) (Credit System) Degree Examinations**

January 2023

19EEE121 - RENEWABLE ENERGY SOURCES

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.*

Unit – I		Marks	BT*	CO*	PO*
1. a)	List the differences between Renewable and Nonrenewable energy sources.	8	L*4	1	1
b)	Explain the advantages and disadvantages of renewable energy.	8	L2	1	1
c)	Explain the applications of solar energy.	4	L2	2	1,7
2. a)	With a neat diagram explain solar water heater system.	8	L2	2	1,7
b)	Define the following terms related to solar radiation geometry. i) Declination ii) Solar constant iii) Hour angle iv) Local Solar time.	8	L1	1	1
c)	List the Applications of Solar PV System.	4	L2	2	1,7
3. a)	List the Advantages and disadvantages of concentrating collector over flat plate collector.	8	L2	2	1,7
b)	With a neat diagram explain Solar cookers.	6	L1	2	1,7
c)	Determine the local solar time and declination at a location latitude 23°15'N, longitude 77°30'E at 12.30 EST on June 19, Equation of time correction is 01'01'', IST = 82.5°E.	6	L2	1	1,7
Unit – II					
4. a)	With a block diagram explain the basic components of Wind Energy conversion system.	8	L2	3	1,2
b)	With a neat diagram explain KVIC Biogas plant.	6	L2	4	1,2
c)	List the classification of wind energy conversion systems.	6	L2	3	1,2
5. a)	Explain the factors to be considered for selection of site for WECS.	8	L2	3	1,7
b)	Explain the factors affecting biogas generation.	8	L2	4	1,7
c)	List the problems associated with wind energy.	4	L1	3	1,7
6. a)	Derive an expression for power available in the wind.	6	L2	3	1,7
b)	Explain the different types of the solar energy storage system.	8	L2	3	1,7
c)	Explain the process of biogas generation.	6	L2	4	1,7
Unit – III					
7. a)	With a neat diagram explain the ocean thermal energy power conversion system.	6	L2	5	1,2,7
b)	Show that energy and power density in the ocean waves is $\frac{P}{A} = \frac{1}{2} g \rho a^2 f$	6	L2	5	1,2,7
c)	List the advantages and limitations of tidal power plant.	8	L2	5	1,2,7
8. a)	Write short notes on i) Hydrogen Energy ii) Wave energy	10	L2	5	1,2,7
b)	Explain Claude cycle of OTEC Power generation.	5	L2	5	1,7
c)	Show that average power per unit basin area for simple single basin tidal system is $0.225R^2$	5	L4	5	1,7

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
